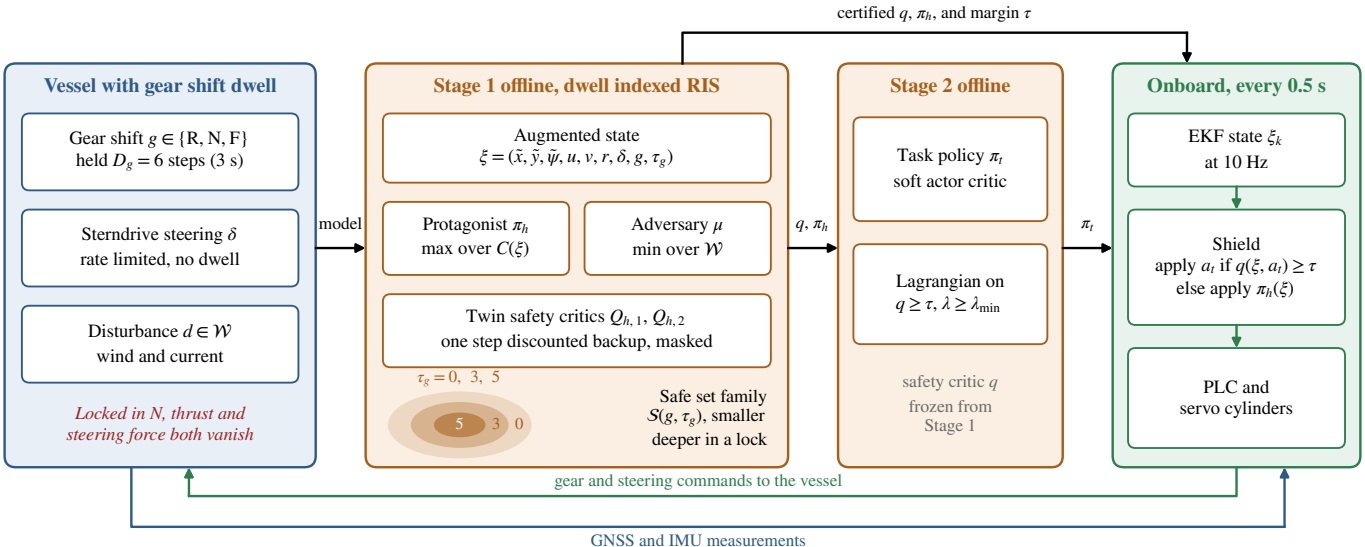
certified q, π_h, and margin τ
Vessel with gear shift dwell
Gear shift $g \in \{R, N, F\}$
held $D_g = 6$ steps (3 s)
Sterndrive steering δ
rate limited, no dwell
Disturbance $d \in \mathcal{W}$
wind and current
Locked in N, thrust and steering force both vanish
model
Stage 1 offline, dwell indexed RIS
Augmented state
$\xi = (\tilde{x}, \tilde{y}, \tilde{\psi}, u, v, r, \delta, g, \tau_g)$
Protagonist π_h
max over $C(\xi)$
Adversary μ
min over $\mathcal{W}$
Twin safety critics $Q_{h,1}$, $Q_{h,2}$
one step discounted backup, masked
$\tau_g = 0,\ 3,\ 5$
5
3
0
Safe set family
$\mathcal{S}(g, \tau_g)$, smaller
deeper in a lock
q, π_h
Stage 2 offline
Task policy π_t
soft actor critic
Lagrangian on
$q \geq \tau$, $\lambda \geq \lambda_{\min}$
safety critic q
frozen from
Stage 1
π_t
Onboard, every 0.5 s
EKF state ξ_k
at 10 Hz
Shield
apply a_t if $q(\xi, a_t) \geq \tau$
else apply $\pi_h(\xi)$
PLC and
servo cylinders
gear and steering commands to the vessel
GNSS and IMU measurements